

STensor/DiffForm

Setup

```
In[*]:= << mGRG`STensor`

In[*]:= $PreRead = ReplaceAll[#,
    expr_String => StringReplace[expr, "SP" -> "mGRG`STensor`Private"]] &;

In[*]:= DefKind[Greek, Alphabet["Greek"]];
DefKind[Capital, ToUpperCase /@ Alphabet[]];
```

Define Differential Forms

DiffFormQ

```
In[*]:= DiffFormQ::usage
Out[*]:= DiffFormQ[name] returns True if 'name'
    is a defined differential form, and False otherwise.
```

인자로 주어진 이름이 DefForm (또는 Fdefine)으로 정의된 것인가를 판단하는 질의함수이다.

```
In[*]:= Fdefine[f, 1];
DiffFormQ /@ {f, A}

Out[*]:= {True, False}
```

DefForm, Fdefine

```
In[*]:= Fdefine::usage
Out[*]:= Fdefine is an alias for DefForm.
```

```
In[*]:= DefForm::usage
Out[*]:= DefForm[f, p, opts] defines a p-form f.
DefForm[f[indices...], p, sym, opts] defines a tensor-valued
    p-form with index symmetries given by the string 'sym'. The
    option PrintAs -> "str" can be used to set the output string.
```

Fdefine은 DefForm의 별칭이다.

새로운 IndexedObject인 미분 형식(Differential Form)을 정의한다. 인자로써는 미분 형식의 이름, 차수(degree), 옵션으로 대칭 문자열이 필요하고, 출력을 위한 문자열이 있다. 출력을 위한 문자열이 없으면 미분 형식의 이름이 출력을 위한 문자열로 사용된다. 또한, 대칭 문자열이 없으면 인덱스가 없는 보통의 미분 형식이 된다.

zero rank

인덱스 랭크가 0이고 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[α, 1];
SP`defineForm[α, 1, "", {}, {}];
{α, α[], α[la], α[la, lb]}

Out[*]= {α, α, αa, α[la, lb]}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[α]]
Out[*]= {0, GenSet[], {Latin}, {}, 1}
```

non-zero rank

랭크가 0이 아닌 미분 형식을 정의하려면 대칭 문자열을 입력해야 한다.

랭크 2이고 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[α[la, lb], 1];
{α, α[la], α[la, lb], α[lμ, la, lb], α[la, lb, lc, ld]}

Out[*]= {α, α[la], αab, αμab, α[la, lb, lc, ld]}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[α]]
Out[*]= {2, Nosymmetric, {Latin, Latin, Latin}, {-1}, 1}
```

랭크 3, 완전 대칭이고 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la, lb, lc], 1, "3+", PrintAs -> "α"];
{A, A[], A[la], A[la, lb], A[lb, la, lc],
 A[ld, la, lb, lc], A[la, lb, lc, ld, le]}

Out[*]= {A, A[], A[la], A[la, lb], αbac, αdabc, A[la, lb, lc, ld, le]}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
Out[*]= {3, Symmetric, {Latin, Latin, Latin, Latin}, {-1}, 1}
```

랭크 3, 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la, lb, uA], 1, PrintAs → "α"];
        {A[la, uμ], A[la, lb, uA], A[lc, la, lb, uA]}

Out[*]:= {A[la, uμ], αabA, αcabA}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]

Out[*]:= {3, Nosymmetric, {Latin, Latin, Latin, Capital}, {-1, -1, 1}, 1}
```

랭크 2, 완전 반대칭, 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la, lb], 1, "2-", PrintAs → "α"];
        {A[lb, la], A[lc, lb, la]}

Out[*]:= {αba, αcba}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]

Out[*]:= {2, Antisymmetric, {Latin, Latin, Latin}, {-1}, 1}
```

랭크 2이고 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[ua, ub], 1, PrintAs → "α"];
        {A[ua, ub], A[lc, ua, ub]}

Out[*]:= {αab, αcab}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]

Out[*]:= {2, Nosymmetric, {Latin, Latin, Latin}, {1}, 1}
```

(도입할 특별한 이유가 없어 보이는) 임의의 랭크인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la], 1, "*", PrintAs → "α"];
        {A, A[], A[lb], A[lc, la], A[lc, la, lb]}

Out[*]:= {A, α, αb, αca, αcab}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]

Out[*]:= {-1, Nosymmetric, {Latin, Latin}, {-1}, 1}
```

(도입할 특별한 이유가 없어 보이는) 임의의 랭크, 완전 반대칭, 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la], 1, "-*", PrintAs -> "α"];
{A, A[], A[la], A[lμ, la], A[lμ, la, lb]}
```

```
Out[*]=
```

```
{A, α, αa, αμa, αμab}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```

```
Out[*]=
```

```
{-1, Antisymmetric, {Latin, Latin}, {-1}, 1}
```

랭크 2, 반대칭, 차수가 1인 미분 형식을 정의한다:

```
In[*]:= Fdefine[A[la, ub], 1, "-ba"]; (* MetricSpaceQ[Latin] == True이므로 문제 없음 *)
{A[lb, ua], A[lc, lb, ua]}
```

```
Out[*]=
```

```
{Aba, Acba}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```

```
Out[*]=
```

```
{2, GenSet[{Cycles[{{1, 2}}], -1}], {Latin, Latin, Latin}, {-1, 1}, 1}
```

```
In[*]:= Fdefine[A[lμ, uv], 1, "-ba"] (* MetricSpaceQ[Greek] == False이므로 문제 있음 *)
```

Msg: incompatible between -ba and {-1, 1}

```
In[*]:= Fdefine[A[lμ, lv], 1, "-ba"];
{A[lv, lμ], A[lα, lv, lμ]}
```

```
Out[*]=
```

```
{Avμ, Aαvμ}
```

```
In[*]:= Through[{GetRank, GetSymmetry, SP`getKindL, SP`getDnupL, SP`getDegree}[A]]
```

```
Out[*]=
```

```
{2, GenSet[{Cycles[{{1, 2}}], -1}], {Latin, Greek, Greek}, {-1}, 1}
```

Check

```
In[*]:= Fdefine[A[la], 1, ""]
```

Msg: incompatible arguments for the length of indices

```
Out[*]=
```

```
$Failed
```

잘못된 인덱스 개수의 확인은 **출력될 때** 자동적으로 진행된다. 그러나 오류 메시지는 프로그램 구조 때문에 출력되지 않는다. 오류 메시지를 출력하려면 SyntaxCheck 함수를 사용한다.

In[*]:= **Fdefine**[f, 0];
{f, f[], f[la]}

Out[*]=
{f, f, f[la]}

In[*]:= **f**[la] // **SyntaxCheck**

Msg: invalid number of indices for f : {la}

Out[*]=
f[la]

In[*]:= **Fdefine**[A, 1]
{A, A[la], A[la, lb], A[la, lb, lc]}

Out[*]=
{A, A_a, A[la, lb], A[la, lb, lc]}

In[*]:= **A**[la, lb] // **SyntaxCheck**

Msg: invalid number of indices for A : {la, lb}

Out[*]=
A[la, lb]

In[*]:= **Fdefine**[A[la], 2];
{A, A[la], A[la, lb], A[la, lb, lc]}

Out[*]=
{A, A_a, A[la, lb], A_{abc}}

In[*]:= **A**[la, lb] // **SyntaxCheck**

Msg: invalid number of indices for A : {la, lb}

Out[*]=
A[la, lb]

In[*]:= **Fdefine**[A[la, lb], 1]
{A, A[la], A[la, lb], A[la, lb, lc]}

Out[*]=
{A, A[la], A_{ab}, A_{abc}}

In[*]:= **A**[la] // **SyntaxCheck**

Msg: invalid number of indices for A : {la}

Out[*]=
A[la]

심볼 형태의 입력에 대해 (Indexed Object에 대한 오류를 확인하는) **SyntaxCheck** 함수는 반응하지 않는다:

$In[*]:=$	A // SyntaxCheck
$Out[*]=$	A
$In[*]:=$	A[]
$Out[*]=$	A[]
$In[*]:=$	A[] // SyntaxCheck
	... Msg: invalid number of indices for A
$Out[*]=$	A[]

UndefForm

In[*]:=

Out[*]=

? A

Symbol

Global`A

UpValue Definitions

mGRG`STensor`Private`getDegree[A] ^= 1

mGRG`STensor`Private`getDnupL[A] ^= {-1}

mGRG`STensor`Private`getGenSet[A] ^= Nosymmetric

mGRG`STensor`Private`getKindL[A] ^= {Latin, Latin, Latin}

mGRG`STensor`Private`getPrtStr[A] ^= A

mGRG`STensor`Private`getRank[A] ^= 2

mGRG`STensor`Private`getType[A] ^= DiffFormQ

IndexedObjectQ[A] ^= True

IndexedOperandQ[A] ^= True

Format Definitions

A /: MakeBoxes[A, StandardForm] := mGRG`STensor`Private`makeFormBox[A, 1, "A

A /: MakeBoxes[A[mGRG`STensor`Private`args\$\_\_\_], StandardForm] := mGRG`STens

Full Name Global`A

^

정의된 미분 형식을 취소한다:

In[*]:=

UndefForm[A]

In[*]:=

DiffFormQ[A]

Out[*]=

False

```
In[*]:= ? A
Out[*]=
```

Symbol
Global`A
Full Name Global`A
^

Differential Form Operators

XP: Exterior Product

XP type의 exterior product 연산자이다.

```
In[*]:= Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]
Out[*]=
```

0-form은 자동적으로 XP에서 제거되고, 또한 순서도 자동적으로 조정된다.

```
In[*]:= {XP[f, f], XP[f, α], XP[β, α], XP[α, α], XP[β, β]}
Out[*]=
```

$$\{f^2, f\alpha, \alpha\wedge\beta, 0, \beta\wedge\beta\}$$

Wedge로 입력할 수 있다. (참고: Wedge를 입력하려면 "`ESC`", `^`, "`ESC`")

```
In[*]:= {f ^ f, f ^ α, β ^ α, α ^ α, β ^ β}
Out[*]=
```

$$\{f^2, f\alpha, \alpha\wedge\beta, 0, \beta\wedge\beta\}$$

DefaultKind의 공간 차원이 고려된다:

```
In[*]:= SetDimension[2]
Out[*]=
```

```
In[*]:= {α ^ f, α ^ β}
Out[*]=
```

$$\{f\alpha, 0\}$$

```
In[*]:= ClearDimension[] (* default *)
```

Indexed DiffForm도 마찬가지이다.

```
In[*]:= Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]; Fdefine[γ[la], 3]
Out[*]=
```

```
In[*]:= {f[la] ^ α[lb], β[la] ^ α[lb], α[lb] ^ α[la], β[la] ^ β[lb]}
Out[*]=
```

$$\{f_a \alpha_b, \alpha_b \wedge \beta_a, -\alpha_a \wedge \alpha_b, \beta_a \wedge \beta_b\}$$

Indexd form에 대한 자동적인 단순화는 이루어지지 않는다:

In[*]:= $\alpha[ua] \wedge \alpha[la]$

Out[*]= $-\alpha_a \wedge \alpha^a$

In[*]:= **SetDimension[2]**

In[*]:= $\{\alpha[la] \wedge f[lb], \alpha[lb] \wedge \beta[la]\}$

Out[*]= $\{f_b \alpha_a, 0\}$

In[*]:= $(\alpha[la] + f[la]) \wedge \beta[ua]$

Out[*]= $f_a \beta^a$

In[*]:= **ClearDimension[] (* default *)**

FtoC 함수의 재귀적 동작을 위해 단순화 규칙 'XP[pre___, XP[args___], post___] := XP[pre, args, post]'은 적용되지 않음:

In[*]:= $\alpha[la] \wedge (\beta[lb] \wedge \gamma[lc])$

Out[*]= $\alpha_a \wedge \beta_b \wedge \gamma_c$

In[*]:= **% // FullForm**

Out[*]//FullForm= $XP[\backslash[Alpha][la], XP[\backslash[Beta][lb], \backslash[Gamma][lc]]]$

XP 표현을 단순화하기 위한 규칙을 따로 설정 할 수 있다:

In[*]:= **simpRules = {XP[pre___, XP[args___], post___] =>
 XP[pre, args, post] /; FreePatternQ[{pre, args, post}]};
 $\alpha[la] \wedge (\beta[lb] \wedge \gamma[lc]) /. \text{simpRules}$
% // FullForm**

Out[*]= $\alpha_a \wedge \beta_b \wedge \gamma_c$

Out[*]//FullForm= $XP[\backslash[Alpha][la], \backslash[Beta][lb], \backslash[Gamma][lc]]$

IP: Interior Product

In[*]:= **IP::usage**

Out[*]=

IP[vector, form] represents the interior product (or interior derivative) of a differential form 'form' with a vector 'vector'. The result is a form of one degree lower.

LD type의 interior product 연산자이다.

In[*]:= **Tdefine[v, 1];**

Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]

In[*]:= **IP[v, #] & /@ {f, α, β, γ}**

Out[*]=

{ $L_v \mathbf{f}$, $L_v \alpha$, $L_v \beta$, $L_v \gamma$ }

$$L_v L_v \omega = 0$$

{IP[v, IP[v, α]], IP[v, IP[v, β]], IP[v, IP[v, γ]]}

Out[*]=

{ $L_v L_v \alpha$, 0, 0}

0-form과 $p \geq 1$ 인 p -form의 곱인 경우 0-form은 자동적으로 IP에서 제거된다:

In[*]:= {IP[v, f], IP[v, f α]}

Out[*]=

{ $L_v \mathbf{f}$, $\mathbf{f} L_v \alpha$ }

선형 연산자이다:

In[*]:= **IP[v, a XD[α] + b β]**

Out[*]=

$b L_v \beta + a L_v d\alpha$

공식:

$$L_v \alpha \wedge \beta = (L_v \alpha) \wedge \beta + (-1)^p \alpha \wedge (L_v \beta)$$

In[*]:= **IP[v, #] & /@ {f ∧ f, f ∧ α, β ∧ α, β ∧ β}**

Out[*]=

{ $L_v \mathbf{f}^2$, $\mathbf{f} L_v \alpha$, $\beta L_v \alpha - \alpha \wedge L_v \beta$, $2 L_v \beta \wedge \beta$ }

XD: Exterior Derivative

XD type의 exterior derivative 연산자이다.

In[*]:= **Fdefine[f, 0]; Fdefine[α, 1]; Fdefine[β, 2]; Fdefine[γ, 3]**

선형 미분 연산자이다.

```
In[*]:= {XD[f], XD[-α], XD[a β]}
Out[*]=
```

```
{df, -dα, a dβ + da^β}
```

Indexed DiffForm에 대해서도 마찬가지이다.

```
In[*]:= Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]; Fdefine[γ[la], 3]
```

```
In[*]:= {XD[f[la]], XD[-α[la]], XD[a β[la]]}
Out[*]=
```

```
{df_a, -dα_a, a dβ_a + da^β_a}
```

```
In[*]:= XD[β[la] ^ α[ua]]
Out[*]=
```

```
dα^a β_a - α^a dβ_a
```

```
In[*]:= XD[α[la] ^ α[ua]]
Out[*]=
```

```
-α_a dα^a + α^a dα_a
```

Power, Log와 같은 ScalarFunction의 XD:

```
In[*]:= Fdefine[f, 0]
```

```
In[*]:= XD[f^2]
Out[*]=
```

```
df^2
```

```
In[*]:= XD[Log[f]]
% // FullForm
Out[*]=
```

```
dLog[f]
```

```
Out[*]//FullForm=
```

```
XD[Log[f]]
```

```
In[*]:= XD[Tscalar[f]]
% // FullForm
Out[*]=
```

```
df
```

```
Out[*]//FullForm=
```

```
XD[f]
```

LD: Lie Derivative

텐서를 위한 LD 연산자는 미분 형식에 대해서도 동일하게 작용한다:

In[*]:=	<code>Tdefine[v, 1]; Fdefine[f[la], 0]; Fdefine[α[la], 1]; Fdefine[β[la], 2]</code>
In[*]:=	<code>{LD[v, f[la]], LD[v, α[ua]], LD[v, β[1]]}</code>
Out[*]=	$\{\mathcal{L}_v \mathbf{f}_a, \mathcal{L}_v \alpha^a, \mathcal{L}_v \beta^1\}$
In[*]:=	<code>{LD[v, α[la] ^ β[lb]], LD[v, XD[α[la]]]}</code>
Out[*]=	$\{\mathcal{L}_v \alpha_a \wedge \beta_b + \alpha_a \wedge \mathcal{L}_v \beta_b, \mathcal{L}_v d\alpha_a\}$

선형 연산자이다.

In[*]:=	<code>SetAttributes[{c1, c2}, Constant]</code>
In[*]:=	<code>LD[v, c1 XD[α[la]] + c2 β[la]]</code>
Out[*]=	$c1 \mathcal{L}_v d\alpha_a + c2 \mathcal{L}_v \beta_a$

LDtoXDRule

공식:

$$\mathcal{L}_v \omega = d(\mathcal{L}_v \omega) + \mathcal{L}_v d\omega$$

In[*]:=	<code>Tdefine[ξ[ua]]; Fdefine[ω[ua], 2]</code>
In[*]:=	<code>LD[ξ, ω[ua]] % /. LDtoXDRule[]</code>
Out[*]=	$\mathcal{L}_\xi \omega^a$
Out[*]=	$\mathcal{L}_\xi d\omega^a + d\mathcal{L}_\xi \omega^a$

HodgeStar

In[*]:=	<code>HodgeStar::usage</code>
Out[*]=	<code>HodgeStar[pForm]</code> computes the Hodge dual of a p-form, resulting in an (n-p)-form, where n is the dimension of the manifold.

```
In[*]:= Tdefine[v, 1];
         Fdefine[f, 0]; Fdefine[a, 1]; Fdefine[b, 2];
         Fdefine[α[la], 1];
         Fdefine[β[la], 1];
         Fdefine[ω[ub], 2];
         Fdefine[Ω[ua, ub], 2, "-ba"]
```

```
In[*]:= GetDimension[DefaultKind] = n; GetSig[DefaultKind] = s;
```

```
In[*]:= HodgeStar /@ {f, a, b}
```

```
Out[*]=
```

```
{*f, *a, *b}
```

```
In[*]:= HodgeStar /@ {α[la], β[la], ω[ub], Ω[la, lb]}
```

```
Out[*]=
```

```
{*αa, *βa, *ωb, *Ωab}
```

공식: with GetSig = s

$$**\omega = (-1)^{s+p(n+1)} \omega, \quad *^{-1} \equiv (-1)^{s+p(n+1)} *$$

```
In[*]:= HodgeStar /@ HodgeStar /@ {α[la], β[la], ω[ub], Ω[la, lb]}
```

```
Out[*]=
```

```
{(-1)1+n+s αa, (-1)1+n+s βa, (-1)2(1+n)+s ωb, (-1)2(1+n)+s Ωab}
```

```
In[*]:= DegreeForm::usage
```

```
Out[*]=
```

DegreeForm[expr] returns the degree of a differential form expression. It correctly computes the degree for base forms, exterior products (XP), exterior derivatives (XD), interior products (IP), Hodge duals (HodgeStar), and codifferentials (CoXD).

```
In[*]:= DegreeForm /@ (HodgeStar /@ {α[la], β[la], Ω[la, lb]})
```

```
Out[*]=
```

```
{-1 + n, -1 + n, -2 + n}
```

```
In[*]:= DegreeForm /@ (HodgeStar /@ {ω[la, lb, uc], ω[ub]})
```

```
Out[*]=
```

```
{n, -2 + n}
```

```
In[*]:= SetDimension[2]
```

In[*]:= ZeroDegreeQ::usage

Out[*]=

ZeroDegreeQ[expr] returns True if 'expr' is a 0-form (a scalar) or not a differential form expression, and False otherwise.

In[*]:= ZeroDegreeQ /@ (HodgeStar /@ {α[la], β[la], Ω[la, lb]})

Out[*]=

{False, False, True}

In[*]:= ZeroDegreeQ /@ (HodgeStar /@ {ω[la, lb, uc], ω[ub]})

Out[*]=

{False, True}

In[*]:= ClearDimension[]

In[*]:= ZeroDegreeQ /@ (HodgeStar /@ {α[la], β[la], ω[ua], Ω[la, lb]})

Out[*]=

{False, False, False, False}

CoXD

In[*]:= CoXD::usage

Out[*]=

CoXD[pForm] represents the codifferential operator acting on a p-form.

In[*]:= Tdefine[v, 1]; Fdefine[f, 0]; Fdefine[A, 1]; Fdefine[B, 2];
Fdefine[α[la], 1];
Fdefine[β[la], 1];
Fdefine[ω[ub], 2];
Fdefine[Ω[ua, ub], 2, "-ba"]

공식: s is the number of minuses in the signature of metric

$$\delta\omega = (-1)^{s+p} n \star d \star \omega = (-1)^p \star^{-1} d \star, \quad \delta\delta = 0$$

In[*]:= GetDimension[DefaultKind] = n; GetSig[DefaultKind] = s;

In[*]:= CoXD[ω[ua]]
% /. CoXDRule[]

Out[*]=

$\delta\omega^a$

Out[*]=

$(-1)^{2n+s} \star d \star \omega^a$

In[*]:= CoXD /@ {f, A, B}

Out[*]=

{0, δA, δB}

In[*]:= CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]}

Out[*]=

{δ_{α_a}, δ_{β_a}, δ_{ω_a}, δ_{Ω_{ab}}}

In[*]:= CoXD /@ CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]}

Out[*]=

{0, 0, 0, 0}

In[*]:= DegreeForm /@ (CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]})

Out[*]=

{0, 0, 1, 1}

In[*]:= ZeroDegreeQ /@ (CoXD /@ {α[la], β[la], ω[la], Ω[la, lb]})

Out[*]=

{True, True, False, False}

Differential Form Operations

ApplyXD

In[*]:= ApplyXD::usage

Out[*]=

ApplyXD[expr] explicitly applies the exterior derivative XD to an expression by expanding it in terms of basis derivatives BD with respect to the defined coordinates.

In[*]:= $\text{expr} = \left\{ \left(1 - 2 \frac{m}{r}\right)^{1/2} \text{XD}[t], \left(1 - 2 \frac{m}{r}\right)^{-1/2} \text{XD}[r], r \text{XD}[\theta], r \sin[\theta] \text{XD}[\phi] \right\}$
% // ApplyXD

Out[*]=

$\left\{ \sqrt{1 - \frac{2m}{r}} dt, \frac{dr}{\sqrt{1 - \frac{2m}{r}}}, r d\theta, r \sin[\theta] d\phi \right\}$

Out[*]=

$\left\{ d\sqrt{1 - \frac{2m}{r}} \wedge dt, d\frac{1}{\sqrt{1 - \frac{2m}{r}}} \wedge dr, dr \wedge d\theta, \sin[\theta] dr \wedge d\phi - r d\phi \wedge d\sin[\theta] \right\}$

```
In[*]:= SetCoordinates[{t, r,  $\theta$ ,  $\phi$ ];
SetAttributes[{m}, Constant]
```

```
In[*]:= expr // ApplyXD
```

```
Out[*]=
```

$$\left\{ -\frac{m \hat{\partial}_r r \, dt \wedge dr}{\sqrt{1 - \frac{2m}{r}} r^2} - \frac{m \hat{\partial}_\theta r \, dt \wedge d\theta}{\sqrt{1 - \frac{2m}{r}} r^2} - \frac{m \hat{\partial}_\phi r \, dt \wedge d\phi}{\sqrt{1 - \frac{2m}{r}} r^2}, \right. \\ \left. \frac{m \hat{\partial}_\theta r \, dr \wedge d\theta}{\left(1 - \frac{2m}{r}\right)^{3/2} r^2} + \frac{m \hat{\partial}_\phi r \, dr \wedge d\phi}{\left(1 - \frac{2m}{r}\right)^{3/2} r^2} - \frac{m \hat{\partial}_t r \, dt \wedge dr}{\left(1 - \frac{2m}{r}\right)^{3/2} r^2}, \hat{\partial}_r r \, dr \wedge d\theta + \hat{\partial}_t r \, dt \wedge d\theta - \hat{\partial}_\phi r \, d\theta \wedge d\phi, \right. \\ \left. r \hat{\partial}_r \theta \cos[\theta] \, dr \wedge d\phi + \hat{\partial}_r r \sin[\theta] \, dr \wedge d\phi + r \hat{\partial}_t \theta \cos[\theta] \, dt \wedge d\phi + \right. \\ \left. \hat{\partial}_t r \sin[\theta] \, dt \wedge d\phi + r \hat{\partial}_\theta \theta \cos[\theta] \, d\theta \wedge d\phi + \hat{\partial}_\theta r \sin[\theta] \, d\theta \wedge d\phi \right\}$$

```
In[*]:= On[EvaluateBDFlag]
```

```
In[*]:= expr // ApplyXD
```

```
Out[*]=
```

$$\left\{ -\frac{m \, dt \wedge dr}{\sqrt{1 - \frac{2m}{r}} r^2}, 0, dr \wedge d\theta, \sin[\theta] \, dr \wedge d\phi + r \cos[\theta] \, d\theta \wedge d\phi \right\}$$

기본 설정으로 복귀한다:

```
In[*]:= Off[EvaluateBDFlag]
```

```
In[*]:= ClearCoordinates[]
```

CollectForm

```
In[*]:= CollectForm::usage
```

```
Out[*]=
```

CollectForm[expr] collects terms in an expression that have the same differential form part, similar to how Collect works for standard algebraic expressions.

```
In[*]:= Tdefine[ $\xi$ [ua]];
Fdefine[w1, 1]; Fdefine[w2, 2]; Fdefine[x[ua], 0]
```

```
In[*]:= expr = IP[ $\xi$ , XD[w1]]
FtoC[expr, {la}]  $\times$  XD[x[ua]]
```

```
Out[*]=
```

$L_\xi dw1$

```
Out[*]=
```

$$\nabla_a w1_b \, dx^b \, \xi^a - \nabla_a w1_b \, dx^a \, \xi^b$$

In[*]:= SumDum[%, {1, 3}]
CollectForm[%]

Out[*]=

$$-\nabla_2 w_{11} \mathbf{dx}^2 \xi^1 + \nabla_1 w_{12} \mathbf{dx}^2 \xi^1 - \nabla_3 w_{11} \mathbf{dx}^3 \xi^1 + \nabla_1 w_{13} \mathbf{dx}^3 \xi^1 + \nabla_2 w_{11} \mathbf{dx}^1 \xi^2 - \nabla_1 w_{12} \mathbf{dx}^1 \xi^2 - \nabla_3 w_{12} \mathbf{dx}^3 \xi^2 + \nabla_2 w_{13} \mathbf{dx}^3 \xi^2 + \nabla_3 w_{11} \mathbf{dx}^1 \xi^3 - \nabla_1 w_{13} \mathbf{dx}^1 \xi^3 + \nabla_3 w_{12} \mathbf{dx}^2 \xi^3 - \nabla_2 w_{13} \mathbf{dx}^2 \xi^3$$

Out[*]=

$$\mathbf{dx}^3 \left(-\nabla_3 w_{11} \xi^1 + \nabla_1 w_{13} \xi^1 - \nabla_3 w_{12} \xi^2 + \nabla_2 w_{13} \xi^2 \right) + \mathbf{dx}^2 \left(-\nabla_2 w_{11} \xi^1 + \nabla_1 w_{12} \xi^1 + \nabla_3 w_{12} \xi^3 - \nabla_2 w_{13} \xi^3 \right) + \mathbf{dx}^1 \left(\nabla_2 w_{11} \xi^2 - \nabla_1 w_{12} \xi^2 + \nabla_3 w_{11} \xi^3 - \nabla_1 w_{13} \xi^3 \right)$$

In[*]:= CoordRep[expr, 3]

Out[*]=

$$\mathbf{dx}^3 \left(-\nabla_3 w_{11} \xi^1 + \nabla_1 w_{13} \xi^1 - \nabla_3 w_{12} \xi^2 + \nabla_2 w_{13} \xi^2 \right) + \mathbf{dx}^2 \left(-\nabla_2 w_{11} \xi^1 + \nabla_1 w_{12} \xi^1 + \nabla_3 w_{12} \xi^3 - \nabla_2 w_{13} \xi^3 \right) + \mathbf{dx}^1 \left(\nabla_2 w_{11} \xi^2 - \nabla_1 w_{12} \xi^2 + \nabla_3 w_{11} \xi^3 - \nabla_1 w_{13} \xi^3 \right)$$

In[*]:= XP[w1, w2]
 $\frac{1}{3!}$ FtoC[%, {la, lb, lc}] $\times$ XD[x[ua]] $\wedge$ XD[x[ub]] $\wedge$ XD[x[uc]]

Out[*]=

$$w1 \wedge w2$$

Out[*]=

$$\frac{1}{2} w_{1a} w_{2bc} \mathbf{dx}^a \wedge \mathbf{dx}^b \wedge \mathbf{dx}^c$$

In[*]:= SumDum[%, {1, 3}] // TindexSort
% // CollectForm

Out[*]=

$$w_{11} w_{223} \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3 - w_{12} w_{213} \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3 + w_{13} w_{212} \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

Out[*]=

$$(w_{11} w_{223} - w_{12} w_{213} + w_{13} w_{212}) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

In[*]:= CoordRep[XP[w1, w2], 3]

Out[*]=

$$(w_{11} w_{223} - w_{12} w_{213} + w_{13} w_{212}) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

In[*]:= RemoveIndexedObject[x]

DegreeForm and ZeroDegreeQ

In[*]:= **DegreeForm::usage**

Out[*]=

DegreeForm[expr] returns the degree of a differential form expression. It correctly computes the degree for base forms, exterior products (XP), exterior derivatives (XD), interior products (IP), Hodge duals (HodgeStar), and codifferentials (CoXD).

In[*]:= **Fdefine**[ω [ua], 2]

In[*]:= { ω [ua], ω [lb, ua], ω [lb, lc, ua]}

DegreeForm /@ %

ZeroDegreeQ /@ %%

Out[*]=

{ ω^a , ω [lb, ua], ω_{bc}^a }

Out[*]=

{2, 0, 0}

Out[*]=

{False, True, True}

FtoC

In[*]:= **FtoC::usage**

Out[*]=

FtoC[expr] or FtoC[expr, {indices}]
converts a differential form expression 'expr' into its corresponding antisymmetric indexed-tensor representation.

In[*]:= **Tdefine**[f[]]; **Tdefine**[ξ [ua]]; **Tdefine**[T[la, lb]];

Fdefine[w0, 0];

Fdefine[w1, 1];

Fdefine[w2, 2];

Fdefine[w3, 3];

Fdefine[w4, 4];

Fdefine[α [la], 1];

Fdefine[β [la], 1];

Fdefine[ω [ua], 2];

Fdefine[Ω [ua, ub], 2, "-ba"]

```
In[*]:= SP`getGenSetOf /@ {ω[ub], ω[la, uc], ω[la, lb, uc]}
SP`getGenSetOf /@ {Ω[la, lb], Ω[la, lb, lc, ld]}
```

```
Out[*]= {GenSet[], GenSet[], GenSet[{Cycles[{1, 2}], -1}]}
```

```
Out[*]= {GenSet[{Cycles[{1, 2}], -1}],
  GenSet[{Cycles[{1, 2}], -1}, {Cycles[{3, 4}], -1}]}
```

```
In[*]:= ω@@@ {{la}, {la, lb}, {la, lb, lc}, {la, lb, lc, ld}}
```

```
Out[*]= {ωa, ω[la, lb], ωabc, ω[la, lb, lc, ld]}
```

p-form

```
In[*]:= f T[la, lb]
% // FtoC
```

```
Out[*]= f Tab
```

```
Out[*]= f Tab
```

```
In[*]:= ξ[ua] × T[lb, lc] w1
% // FtoC
```

```
Out[*]= w1 Tbc ξa
```

```
Out[*]= Tbc w1d ξa
```

```
In[*]:= T[la, lb] w1
% // FtoC
```

```
Out[*]= w1 Tab
```

```
Out[*]= Tab w1c
```

```
In[*]:= T[la, lb] × ω[uc]
% // FtoC
```

```
Out[*]= Tab ωc
```

```
Out[*]= Tab ωdec
```

<code>In[*]:=</code>	<code>T[la, lb] w1 ^ w2</code> <code>% // FtoC</code>
<code>Out[*]=</code>	$T_{ab} w1^a w2^b$
<code>Out[*]=</code>	$T_{ab} w1_e w2_{cd} - T_{ab} w1_d w2_{ce} + T_{ab} w1_c w2_{de}$
<code>In[*]:=</code>	<code>ω[ua]</code> <code>% // FtoC</code>
<code>Out[*]=</code>	ω^a
<code>Out[*]=</code>	ω_{bc}^a
<code>In[*]:=</code>	<code>T[la, lb] w1 ^ ω[uc]</code> <code>% // FtoC</code>
<code>Out[*]=</code>	$T_{ab} w1^a \omega^b$
<code>Out[*]=</code>	$T_{ab} w1_f \omega_{de}^c - T_{ab} w1_e \omega_{df}^c + T_{ab} w1_d \omega_{ef}^c$
<code>In[*]:=</code>	<code>Ω[ua, ub]</code> <code>% // FtoC</code>
<code>Out[*]=</code>	Ω^{ab}
<code>Out[*]=</code>	Ω_{cd}^{ab}

XD

<code>In[*]:=</code>	<code>Off[XDtoCDfrag]</code>
<code>In[*]:=</code>	<code>XD[x]</code> <code>% // FtoC</code>
<code>Out[*]=</code>	dx
<code>Out[*]=</code>	$\partial_a x$
<code>In[*]:=</code>	<code>On[XDtoCDfrag]</code>

In[*]:= XD[x]
% // FtoC

Out[*]=

dx

Out[*]=

$\nabla_a X$

In[*]:= TorsionFreeQ[CD] = False;

In[*]:= XD[w1]
% // FtoC

Out[*]=

dw1

Out[*]=

$\partial_a w1_b - \partial_b w1_a$

In[*]:= TorsionFreeQ[CD] = True;

In[*]:= XD[w1]
% // FtoC

Out[*]=

dw1

Out[*]=

$\nabla_a w1_b - \nabla_b w1_a$

In[*]:= T[la, lb] w1 ^ XD[w1]
% // FtoC

Out[*]=

$T_{ab} w1^a dw1^b$

Out[*]=

$\nabla_d w1_e T_{ab} w1_c - \nabla_e w1_d T_{ab} w1_c - \nabla_c w1_e T_{ab} w1_d + \nabla_e w1_c T_{ab} w1_d + \nabla_c w1_d T_{ab} w1_e - \nabla_d w1_c T_{ab} w1_e$

In[*]:= T[la, lb] w1 ^ XD[ω[ua]]
% // FtoC

Out[*]=

$T_{ab} w1^a dw1^b$

Out[*]=

$\nabla_d \omega_{ef}^a T_{ab} w1_c - \nabla_e \omega_{df}^a T_{ab} w1_c + \nabla_f \omega_{de}^a T_{ab} w1_c - \nabla_c \omega_{ef}^a T_{ab} w1_d +$
 $\nabla_e \omega_{cf}^a T_{ab} w1_d - \nabla_f \omega_{ce}^a T_{ab} w1_d + \nabla_c \omega_{df}^a T_{ab} w1_e - \nabla_d \omega_{cf}^a T_{ab} w1_e +$
 $\nabla_f \omega_{cd}^a T_{ab} w1_e - \nabla_c \omega_{de}^a T_{ab} w1_f + \nabla_d \omega_{ce}^a T_{ab} w1_f - \nabla_e \omega_{cd}^a T_{ab} w1_f$

```
In[*]:= T[la, lb] × ω[uc] ∧ XD[Ω[ua, ub]]
% // FtoC
```

```
Out[*]=
```

$$T_{ab} \omega^c \wedge d\Omega^{ab}$$

```
Out[*]=
```

$$\begin{aligned} & \nabla_f \Omega_{gh}^{ab} T_{ab} \omega_{de}^c - \nabla_g \Omega_{fh}^{ab} T_{ab} \omega_{de}^c + \nabla_h \Omega_{fg}^{ab} T_{ab} \omega_{de}^c - \nabla_e \Omega_{gh}^{ab} T_{ab} \omega_{df}^c + \nabla_g \Omega_{eh}^{ab} T_{ab} \omega_{df}^c - \\ & \nabla_h \Omega_{eg}^{ab} T_{ab} \omega_{df}^c + \nabla_e \Omega_{fh}^{ab} T_{ab} \omega_{dg}^c - \nabla_f \Omega_{eh}^{ab} T_{ab} \omega_{dg}^c + \nabla_h \Omega_{ef}^{ab} T_{ab} \omega_{dg}^c - \nabla_e \Omega_{fg}^{ab} T_{ab} \omega_{dh}^c + \\ & \nabla_f \Omega_{eg}^{ab} T_{ab} \omega_{dh}^c - \nabla_g \Omega_{ef}^{ab} T_{ab} \omega_{dh}^c + \nabla_d \Omega_{gh}^{ab} T_{ab} \omega_{ef}^c - \nabla_g \Omega_{dh}^{ab} T_{ab} \omega_{ef}^c + \nabla_h \Omega_{dg}^{ab} T_{ab} \omega_{ef}^c - \\ & \nabla_d \Omega_{fh}^{ab} T_{ab} \omega_{eg}^c + \nabla_f \Omega_{dh}^{ab} T_{ab} \omega_{eg}^c - \nabla_h \Omega_{df}^{ab} T_{ab} \omega_{eg}^c + \nabla_d \Omega_{fg}^{ab} T_{ab} \omega_{eh}^c - \nabla_f \Omega_{dg}^{ab} T_{ab} \omega_{eh}^c + \\ & \nabla_g \Omega_{df}^{ab} T_{ab} \omega_{eh}^c + \nabla_d \Omega_{eh}^{ab} T_{ab} \omega_{fg}^c - \nabla_e \Omega_{dh}^{ab} T_{ab} \omega_{fg}^c + \nabla_h \Omega_{de}^{ab} T_{ab} \omega_{fg}^c - \nabla_d \Omega_{eg}^{ab} T_{ab} \omega_{fh}^c + \\ & \nabla_e \Omega_{dg}^{ab} T_{ab} \omega_{fh}^c - \nabla_g \Omega_{de}^{ab} T_{ab} \omega_{fh}^c + \nabla_d \Omega_{ef}^{ab} T_{ab} \omega_{gh}^c - \nabla_e \Omega_{df}^{ab} T_{ab} \omega_{gh}^c + \nabla_f \Omega_{de}^{ab} T_{ab} \omega_{gh}^c \end{aligned}$$

XP

```
In[*]:= XP[XD[x], XD[y]]
% // FtoC
```

```
Out[*]=
```

$$dx \wedge dy$$

```
Out[*]=
```

$$-\nabla_a y \nabla_b x + \nabla_a x \nabla_b y$$

```
In[*]:= XP[w1, w2]
% // FtoC
```

```
Out[*]=
```

$$w1 \wedge w2$$

```
Out[*]=
```

$$w1_c w2_{ab} - w1_b w2_{ac} + w1_a w2_{bc}$$

```
In[*]:= T[la, lb] × XP[w1, w2]
% // FtoC
```

```
Out[*]=
```

$$T_{ab} w1 \wedge w2$$

```
Out[*]=
```

$$T_{ab} w1_e w2_{cd} - T_{ab} w1_d w2_{ce} + T_{ab} w1_c w2_{de}$$

LD

$In[*] :=$	$LD[\xi, w0]$ % // FtoC
$Out[*] =$	$\mathcal{L}_\xi w0$
$Out[*] =$	$\nabla_a w0 \xi^a$
$In[*] :=$	$LD[\xi, w1]$ % // FtoC
$Out[*] =$	$\mathcal{L}_\xi w1$
$Out[*] =$	$\mathcal{L}_\xi w1_a$
$In[*] :=$	$LD[\xi, \omega[ua]]$ % // FtoC
$Out[*] =$	$\mathcal{L}_\xi \omega^a$
$Out[*] =$	$\mathcal{L}_\xi \omega_{bc}^a$
$In[*] :=$	$LD[\xi, \Omega[ua, ub]]$ % // FtoC
$Out[*] =$	$\mathcal{L}_\xi \Omega^{ab}$
$Out[*] =$	$\mathcal{L}_\xi \Omega_{cd}^{ab}$
$In[*] :=$	$LD[\xi, w1 \wedge w2]$ % // FtoC
$Out[*] =$	$w1 \wedge \mathcal{L}_\xi w2 + \mathcal{L}_\xi w1 \wedge w2$
$Out[*] =$	$\mathcal{L}_\xi w2_{bc} w1_a - \mathcal{L}_\xi w2_{ac} w1_b + \mathcal{L}_\xi w2_{ab} w1_c + \mathcal{L}_\xi w1_c w2_{ab} - \mathcal{L}_\xi w1_b w2_{ac} + \mathcal{L}_\xi w1_a w2_{bc}$

In[*]:= **LD[ξ , XD[x]]**
% // FtoC

Out[*]=

$$\mathcal{L}_\xi \mathbf{d}x$$

Out[*]=

$$\mathcal{L}_\xi \nabla_a x$$

Check:

In[*]:= **LD[ξ , XD[x]]**
% /. LDtoXDRule[]
% // FtoC

Out[*]=

$$\mathcal{L}_\xi \mathbf{d}x$$

Out[*]=

$$\mathbf{d} \mathcal{L}_\xi \mathbf{d}x$$

Out[*]=

$$\nabla_a \xi^b \nabla_b x + \nabla_a \nabla_b x \xi^b$$

In[*]:= **LD[ξ , CD[la, x]]**
% // LDtoCD

Out[*]=

$$\mathcal{L}_\xi \nabla_a x$$

Out[*]=

$$\nabla_a \xi^b \nabla_b x + \nabla_b \nabla_a x \xi^b$$

IP

In[*]:= **IP[ξ , w0] (* wrong operation *)**
% // FtoC

Out[*]=

$$\mathcal{L}_\xi \mathbf{w0}$$

Out[*]=

$$\mathcal{L}_\xi \mathbf{w0}$$

In[*]:= **IP[ξ , w1]**
% // FtoC

Out[*]=

$$\mathcal{L}_\xi \mathbf{w1}$$

Out[*]=

$$w1_a \xi^a$$

```
In[*]:= IP[ξ, XP[w1, w2]]
% // FtoC
```

```
Out[*]=
```

$$w2 \mathcal{L}_\xi w1 - w1 \wedge \mathcal{L}_\xi w2$$

```
Out[*]=
```

$$w1_c w2_{ab} \xi^c - w1_b w2_{ac} \xi^c + w1_a w2_{bc} \xi^c$$

```
In[*]:= IP[ξ, XD[w1]]
% // FtoC
```

```
Out[*]=
```

$$\mathcal{L}_\xi dw1$$

```
Out[*]=
```

$$-\nabla_a w1_b \xi^b + \nabla_b w1_a \xi^b$$

```
In[*]:= IP[ξ, XP[XD[x], XD[y]]]
% // FtoC
```

```
Out[*]=
```

$$-\mathcal{L}_\xi dy dx + \mathcal{L}_\xi dx dy$$

```
Out[*]=
```

$$\nabla_a y \nabla_b x \xi^b - \nabla_a x \nabla_b y \xi^b$$

HodgeStar

참고: R. M. Wald (Appendix B.2, Problem 4.2.a)

$$\epsilon^{\mu_1 \dots \mu_p \alpha_1 \dots \alpha_{n-p}} \epsilon_{\mu_1 \dots \mu_p \beta_1 \dots \beta_{n-p}} = (-1)^s p! (n-p)! \delta_{\beta_1}^{[\alpha_1} \dots \delta_{\beta_{n-p}}^{\alpha_{n-p}]}$$

$$(*\alpha)_{\beta_1 \dots \beta_{n-p}} = \frac{1}{p!} \alpha^{\alpha_1 \dots \alpha_p} \epsilon_{\alpha_1 \dots \alpha_p \beta_1 \dots \beta_{n-p}}$$

$$**\alpha = (-1)^{s+p(n-p)} \alpha = (-1)^{s+p(n+1)} \alpha$$

where s is the number of minuses appearing in the signature of g_{ab} .

```
In[*]:= HodgeStar[XD[x]]
% // FtoC
```

```
Out[*]=
```

$$*dx$$

Msg: Need to SetDimension []

```
Out[*]=
```

\$Failed

```
In[*]:= SetDimension[3]; SetSig[0]
```

In[*]:= **HodgeStar[XD[x]]**
% // FtoC

Out[*]=

***dx**

Out[*]=

$\nabla_c X \in^c_{ab}$

In[*]:= **HodgeStar[w1]**
% // FtoC

Out[*]=

***w1**

Out[*]=

$\in^c_{ab} w1_c$

In[*]:= **HodgeStar[ω[ua]]**
% // FtoC

Out[*]=

***ω^a**

Out[*]=

$\frac{1}{2} \in^{cd}_b \omega_{cd}^a$

In[*]:= **HodgeStar[XP[XD[x], XD[y]]]**
% // FtoC

Out[*]=

***dx^dy**

Out[*]=

$-\frac{1}{2} \nabla_b y \nabla_c x \in^{bc}_a + \frac{1}{2} \nabla_b x \nabla_c y \in^{bc}_a$

In[*]:= **HodgeStar[XP[w1, Ω[ub, uc]]]**
% // FtoC

Out[*]=

***w1^Ω^{bc}**

Out[*]=

$\frac{1}{6} \in^{ade} w1_e \Omega_{ad}^{bc} - \frac{1}{6} \in^{ade} w1_d \Omega_{ae}^{bc} + \frac{1}{6} \in^{ade} w1_a \Omega_{de}^{bc}$

In[*]:= **HodgeStar**[LD[ξ, XD[x]]]
% // FtoC

Out[*]=

$$\star \mathcal{L}_\xi \mathbf{d}x$$

Out[*]=

$$\in^c_{ab} \mathcal{L}_\xi \nabla_c X$$

In[*]:= **HodgeStar**[LD[ξ, XD[ω[ua]]]]
% // FtoC

Out[*]=

$$\star \mathcal{L}_\xi \mathbf{d}\omega^a$$

Out[*]=

$$\frac{1}{6} \in^{bcd} \mathcal{L}_\xi \nabla_b \omega_{cd}^a - \frac{1}{6} \in^{bcd} \mathcal{L}_\xi \nabla_c \omega_{bd}^a + \frac{1}{6} \in^{bcd} \mathcal{L}_\xi \nabla_d \omega_{bc}^a$$

In[*]:= **HodgeStar**[IP[ξ, XD[x]]]
% // FtoC

Out[*]=

$$\star \mathcal{L}_\xi \mathbf{d}x$$

Out[*]=

$$\nabla_d X \in_{abc} \xi^d$$

In[*]:= **HodgeStar**[IP[ξ, XD[w1[]]]]
% // FtoC

Out[*]=

$$\star \mathcal{L}_\xi \mathbf{d}w_1$$

Out[*]=

$$\nabla_c w_{1d} \in^d_{ab} \xi^c - \nabla_c w_{1d} \in^c_{ab} \xi^d$$

In[*]:= **HodgeStar**[IP[ξ, XD[ω[ua]]]]
% // FtoC

Out[*]=

$$\star \mathcal{L}_\xi \mathbf{d}\omega^a$$

Out[*]=

$$\frac{1}{2} \nabla_c \omega_{de}^a \in^{de}_b \xi^c + \frac{1}{2} \nabla_c \omega_{de}^a \in^{cd}_b \xi^e - \frac{1}{2} \nabla_c \omega_{de}^a \in^{dc}_b \xi^e$$

In[*]:= **HodgeStar**[IP[ξ , XD[Ω [ua, ub]]]]
% // FtoC

Out[*]=

$\star L_{\xi} d\Omega^{ab}$

Out[*]=

$$\frac{1}{2} \nabla_d \Omega_{ef}^{ab} \in_{ef}^c \xi^d + \frac{1}{2} \nabla_d \Omega_{ef}^{ab} \in_{de}^c \xi^f - \frac{1}{2} \nabla_d \Omega_{ef}^{ab} \in_{ed}^c \xi^f$$

In[*]:= **HodgeStar**[x]
% // FtoC

Out[*]=

$\star X$

Out[*]=

$X \in_{abc}$

In[*]:= **CoXD**[w1] /. **CoXDRule**[]

Out[*]=

$-\star d \star w1$

In[*]:= **-HodgeStar**[w1]
% // FtoC

Out[*]=

$-\star w1$

Out[*]=

$-\in_{ab}^c w1_c$

In[*]:= **-XD**[**HodgeStar**[w1]]
% // FtoC

Out[*]=

$-\star d \star w1$

Out[*]=

$$-\nabla_c w1_d \in_{ab}^d + \nabla_b w1_d \in_{ac}^d - \nabla_a w1_d \in_{bc}^d$$

In[*]:= **-HodgeStar**[XD[**HodgeStar**[w1]]]
% // FtoC

Out[*]=

$-\star d \star w1$

Out[*]=

$$-\frac{1}{6} \nabla_a w1_b \in_{cd}^b \in^{acd} + \frac{1}{6} \nabla_a w1_b \in_{cd}^b \in^{cad} - \frac{1}{6} \nabla_a w1_b \in_{cd}^b \in^{cda}$$

```
In[*]:= % /. EpsilonProductRule[]
% // Absorbg
```

```
Out[*]=
```

$$-\nabla_a w_{1b} g^{ba}$$

```
Out[*]=
```

$$-\nabla^a w_{1a}$$

```
In[*]:= CoXD[w1]
% // FtoC
```

```
Out[*]=
```

$$\delta w_1$$

```
Out[*]=
```

$$-\nabla^a w_{1a}$$

```
In[*]:= ClearDimension[]; ClearSig[]
```

CoXD

```
In[*]:= SetDimension[3]; SetSig[0]
```

```
In[*]:= CoXD[XD[x]]
% /. CoXDRule[]
```

```
Out[*]=
```

$$\delta dx$$

```
Out[*]=
```

$$- *d * dx$$

```
In[*]:= -HodgeStar[XD[HodgeStar[XD[x]]]]
% // FtoC
```

```
Out[*]=
```

$$- *d * dx$$

```
Out[*]=
```

$$-\frac{1}{6} \nabla_a \nabla_b x \in_{cd}{}^b \in^{acd} + \frac{1}{6} \nabla_a \nabla_b x \in_{cd}{}^b \in^{cad} - \frac{1}{6} \nabla_a \nabla_b x \in_{cd}{}^b \in^{cda}$$

```
In[*]:= % /. EpsilonProductRule[]
% // Absorbg
```

```
Out[*]=
```

$$-\nabla_a \nabla_b x g^{ba}$$

```
Out[*]=
```

$$-\nabla^a \nabla_a x$$

In[*]:= **CoXD[XD[x]]**
% // FtoC

Out[*]=

$\delta d x$

Out[*]=

$-\nabla^a \nabla_a X$

In[*]:= **CoXD[w1]**
% // FtoC

Out[*]=

$\delta w1$

Out[*]=

$-\nabla^a w1_a$

In[*]:= **CoXD[w1]**
% /. CoXDRule[]

Out[*]=

$\delta w1$

Out[*]=

$-\star d \star w1$

In[*]:= **-HodgeStar[XD[HodgeStar[w1]]]**
% // FtoC

Out[*]=

$-\star d \star w1$

Out[*]=

$-\frac{1}{6} \nabla_a w1_b \in_{cd}{}^b \in{}^{acd} + \frac{1}{6} \nabla_a w1_b \in_{cd}{}^b \in{}^{cad} - \frac{1}{6} \nabla_a w1_b \in_{cd}{}^b \in{}^{cda}$

In[*]:= **% /. EpsilonProductRule[]**
% // Absorbg

Out[*]=

$-\nabla_a w1_b g^{ba}$

Out[*]=

$-\nabla^a w1_a$

In[*]:= CoXD[w2]
% // FtoC

Out[*]=

δw_2

Out[*]=

$-\nabla^b w_{2ba}$

In[*]:= CoXD[w2]
% /. CoXDRule[]

Out[*]=

δw_2

Out[*]=

$*d*w_2$

In[*]:= HodgeStar[XD[HodgeStar[w2]]]
% // FtoC

Out[*]=

$*d*w_2$

Out[*]=

$\frac{1}{4} \nabla_b w_{cd} \in_e^{cd} \in_a^{be} - \frac{1}{4} \nabla_b w_{cd} \in_e^{cd} \in_a^{eb}$

In[*]:= % /. EpsilonProductRule[]
% // Absorbg
% // TindexSort

Out[*]=

$\frac{1}{2} \nabla_b w_{ac} g^{cb} - \frac{1}{2} \nabla_b w_{ca} g^{cb}$

Out[*]=

$\frac{1}{2} \nabla^b w_{ab} - \frac{1}{2} \nabla^b w_{ba}$

Out[*]=

$\nabla_b w_a^b$

In[*]:= CoXD[w3]
% // FtoC

Out[*]=

δw_3

Out[*]=

$-\nabla^c w_{cab}$

```
In[*]:= CoXD[w3]
% /. CoXDRule[]
```

```
Out[*]=
δw3
```

```
Out[*]=
- *d*w3
```

```
In[*]:= -HodgeStar[XD[HodgeStar[w3]]]
% // FtoC
```

```
Out[*]=
- *d*w3
```

```
Out[*]=
- 1/6 ∇cw3def ∈cab ∈def
```

```
In[*]:= % /. EpsilonProductRule[]
% // Absorbg
% // TindexSort
```

```
Out[*]=
- 1/6 ∇cw3abd gcd + 1/6 ∇cw3adb gcd + 1/6 ∇cw3bad gcd - 1/6 ∇cw3bda gcd - 1/6 ∇cw3dab gcd + 1/6 ∇cw3dba gcd
```

```
Out[*]=
- 1/6 ∇cw3abc + 1/6 ∇cw3acb + 1/6 ∇cw3bac - 1/6 ∇cw3bca - 1/6 ∇cw3cab + 1/6 ∇cw3cba
```

```
Out[*]=
- ∇cw3abc
```

공식:

$$(\delta\omega)_{a\dots b} = -\bar{\nabla}^p \omega_{pa\dots b}, \quad \bar{\nabla}_a : \text{torsion-free CD}$$

Laplace-deRham operator

$$\Delta = -(\delta d + d\delta)$$

```
In[*]:= Δ[expr_] := -(CoXD[XD[expr]] + XD[CoXD[expr]])
```

```
In[*]:= Δ[w0]
% // FtoC
```

```
Out[*]=
- δdw0
```

```
Out[*]=
∇a∇aw0
```

In[*]:= $\Delta[w1]$
% // FtoC

Out[*]=
 $-\delta dw1 - d\delta w1$

Out[*]=
 $\nabla_a \nabla_b w1^b - \nabla^b \nabla_a w1_b + \nabla^b \nabla_b w1_a$

In[*]:= $\text{CoXD}[\omega[ua]]$
% // FtoC

Out[*]=
 $\delta \omega^a$

Out[*]=
 $-\nabla^c \omega_{cb}^a$

In[*]:= $\text{XP}[\text{XD}[x], \text{XD}[y]]$
% // CoXD
% // FtoC

Out[*]=
 $dx \wedge dy$

Out[*]=
 $\delta dx \wedge dy$

Out[*]=
 $\nabla_b y \nabla^b \nabla_a x - \nabla_b x \nabla^b \nabla_a y - \nabla_a y \nabla^b \nabla_b x + \nabla_a x \nabla^b \nabla_b y$

In[*]:= $\text{XP}[w1, \Omega[ub, uc]]$
% // CoXD
% // FtoC

Out[*]=
 $w1 \wedge \Omega^{bc}$

Out[*]=
 $\delta w1 \wedge \Omega^{bc}$

Out[*]=
 $-\nabla^e \Omega_{de}^{bc} w1_a + \nabla^e \Omega_{ae}^{bc} w1_d - \nabla^e \Omega_{ad}^{bc} w1_e - \nabla^e w1_e \Omega_{ad}^{bc} + \nabla^e w1_d \Omega_{ae}^{bc} - \nabla^e w1_a \Omega_{de}^{bc}$

$In[*]:=$ **LD**[ξ , XD[x]]

% // CoXD

% // FtoC

$Out[*]=$

$$\mathcal{L}_\xi dx$$

$Out[*]=$

$$\delta \mathcal{L}_\xi dx$$

$Out[*]=$

$$-\nabla^a \mathcal{L}_\xi \nabla_a x$$

$In[*]:=$ **LD**[ξ , XD[$\omega[ua]$]]

% // CoXD

% // FtoC

$Out[*]=$

$$\mathcal{L}_\xi d\omega^a$$

$Out[*]=$

$$\delta \mathcal{L}_\xi d\omega^a$$

$Out[*]=$

$$-\nabla^d \mathcal{L}_\xi \nabla_b \omega_{cd}^a + \nabla^d \mathcal{L}_\xi \nabla_c \omega_{bd}^a - \nabla^d \mathcal{L}_\xi \nabla_d \omega_{bc}^a$$

$In[*]:=$ **IP**[ξ , XD[x]]

% // CoXD

% // FtoC

$Out[*]=$

$$\mathcal{L}_\xi dx$$

$Out[*]=$

$$0$$

$Out[*]=$

$$0$$

$In[*]:=$ **IP**[ξ , XD[w1]]

% // CoXD

% // FtoC

$Out[*]=$

$$\mathcal{L}_\xi dw1$$

$Out[*]=$

$$\delta \mathcal{L}_\xi dw1$$

$Out[*]=$

$$\nabla_a w1_b \nabla^a \xi^b - \nabla_a w1_b \nabla^b \xi^a + \nabla^a \nabla_a w1_b \xi^b - \nabla^a \nabla_b w1_a \xi^b$$

```
In[*]:= IP[ξ, XD[ω[ua]]]
% // CoXD
% // FtoC
```

```
Out[*]=
```

$$\mathcal{L}_\xi \mathbf{d}\omega^a$$

```
Out[*]=
```

$$\delta \mathcal{L}_\xi \mathbf{d}\omega^a$$

```
Out[*]=
```

$$\nabla_b \omega_{cd}^a \nabla^c \xi^d - \nabla_c \omega_{bd}^a \nabla^c \xi^d + \nabla_c \omega_{bd}^a \nabla^d \xi^c + \nabla^c \nabla_b \omega_{cd}^a \xi^d - \nabla^c \nabla_c \omega_{bd}^a \xi^d + \nabla^c \nabla_d \omega_{bc}^a \xi^d$$

```
In[*]:= IP[ξ, XD[Ω[ua, ub]]]
% // CoXD
% // FtoC
```

```
Out[*]=
```

$$\mathcal{L}_\xi \mathbf{d}\Omega^{ab}$$

```
Out[*]=
```

$$\delta \mathcal{L}_\xi \mathbf{d}\Omega^{ab}$$

```
Out[*]=
```

$$\nabla_c \Omega_{de}^{ab} \nabla^d \xi^e - \nabla_d \Omega_{ce}^{ab} \nabla^d \xi^e + \nabla_d \Omega_{ce}^{ab} \nabla^e \xi^d + \nabla^d \nabla_c \Omega_{de}^{ab} \xi^e - \nabla^d \nabla_d \Omega_{ce}^{ab} \xi^e + \nabla^d \nabla_e \Omega_{cd}^{ab} \xi^e$$

Coordinate Representation: CoordRep

```
In[*]:= CoordRep::usage
```

```
Out[*]=
```

CoordRep[form, coSys] gives the coordinate representation of the differential 'form' using the coordinate system 'coSys'.

CoordRep[form, n] gives the representation in a generic n-dimensional space, with the dimension defaulting to that of the Latin kind.

```
In[*]:= Tdefine[v, 1]; Tdefine[ξ[ua]];
Fdefine[w1, 1];
Fdefine[w2, 2];
Fdefine[x[ua], 0]
```

```
In[*]:= IP[ξ, XD[w1]]
% // FtoC
```

```
Out[*]=
```

$$\mathcal{L}_\xi \mathbf{d}w_1$$

```
Out[*]=
```

$$-\nabla_a w_1{}_b \xi^b + \nabla_b w_1{}_a \xi^b$$

In[*]:= SumDum[% XD[x[ua]], {1, 3}] // CollectForm

Out[*]=

$$\begin{aligned} & \mathbf{dx}^3 \left(-\nabla_3 w_{11} \xi^1 + \nabla_1 w_{13} \xi^1 - \nabla_3 w_{12} \xi^2 + \nabla_2 w_{13} \xi^2 \right) + \\ & \mathbf{dx}^2 \left(-\nabla_2 w_{11} \xi^1 + \nabla_1 w_{12} \xi^1 + \nabla_3 w_{12} \xi^3 - \nabla_2 w_{13} \xi^3 \right) + \\ & \mathbf{dx}^1 \left(\nabla_2 w_{11} \xi^2 - \nabla_1 w_{12} \xi^2 + \nabla_3 w_{11} \xi^3 - \nabla_1 w_{13} \xi^3 \right) \end{aligned}$$

In[*]:= CoordRep[IP[ξ, XD[w1]], 3]

Out[*]=

$$\begin{aligned} & \mathbf{dx}^3 \left(-\nabla_3 w_{11} \xi^1 + \nabla_1 w_{13} \xi^1 - \nabla_3 w_{12} \xi^2 + \nabla_2 w_{13} \xi^2 \right) + \\ & \mathbf{dx}^2 \left(-\nabla_2 w_{11} \xi^1 + \nabla_1 w_{12} \xi^1 + \nabla_3 w_{12} \xi^3 - \nabla_2 w_{13} \xi^3 \right) + \\ & \mathbf{dx}^1 \left(\nabla_2 w_{11} \xi^2 - \nabla_1 w_{12} \xi^2 + \nabla_3 w_{11} \xi^3 - \nabla_1 w_{13} \xi^3 \right) \end{aligned}$$

In[*]:= XP[w1, w2]

% // FtoC

Out[*]=

$$\mathbf{w1} \wedge \mathbf{w2}$$

Out[*]=

$$w_{1c} w_{2ab} - w_{1b} w_{2ac} + w_{1a} w_{2bc}$$

In[*]:= $\frac{1}{3!}$ % XD[x[ua]] ^ XD[x[ub]] ^ XD[x[uc]]

Out[*]=

$$\frac{1}{2} w_{1a} w_{2bc} \mathbf{dx}^a \wedge \mathbf{dx}^b \wedge \mathbf{dx}^c$$

In[*]:= SumDum[%, {1, 3}] // TindexSort // CollectForm

Out[*]=

$$\left(w_{11} w_{223} - w_{12} w_{213} + w_{13} w_{212} \right) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

In[*]:= CoordRep[XP[w1, w2], 3]

Out[*]=

$$\left(w_{11} w_{223} - w_{12} w_{213} + w_{13} w_{212} \right) \mathbf{dx}^1 \wedge \mathbf{dx}^2 \wedge \mathbf{dx}^3$$

In[*]:= SetDimension[3]

In[*]:= Fdefine[ω[ua], 2]

In[*]:= $\omega[ua]$
% // FtoC

Out[*]=

$$\omega^a$$

Out[*]=

$$\omega_{bc}^a$$

In[*]:= $\frac{1}{\text{DegreeForm}[\omega[ua]]}$ % XP[XD[x[ub]], XD[x[uc]]]

Out[*]=

$$\frac{1}{2} \mathbf{dx}^b \wedge \mathbf{dx}^c \omega_{bc}^a$$

In[*]:= % // SumDum // TindexSort

Out[*]=

$$\mathbf{dx}^2 \wedge \mathbf{dx}^3 \omega_{23}^a + \mathbf{dx}^1 \wedge \mathbf{dx}^3 \omega_{13}^a + \mathbf{dx}^1 \wedge \mathbf{dx}^2 \omega_{12}^a$$

In[*]:= CoordRep[$\omega[ua]$]

Out[*]=

$$\mathbf{dx}^2 \wedge \mathbf{dx}^3 \omega_{23}^a + \mathbf{dx}^1 \wedge \mathbf{dx}^3 \omega_{13}^a + \mathbf{dx}^1 \wedge \mathbf{dx}^2 \omega_{12}^a$$